SoK: Market Microstructure for Decentralized Prediction Markets (DePMs)*

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Abstract. Decentralized prediction markets (DePMs) allow open participation in event-based wagering without fully relying on centralized intermediaries. We review the history of DePMs which date back to 2011 and includes hundreds of proposals. Perhaps surprising, modern DePMs like Polymarket deviate materially from earlier designs like Truthcoin and Augur v1. We use our review to present a modular workflow comprising eight stages: underlying infrastructure, market topic, share structure and pricing, initialization, trading, market resolution, settlement, and archiving. For each module, we enumerate the design variants, analyzing trade-offs around decentralization, expressiveness, and manipulation resistance. We also identify open problems for researchers interested in this ecosystem.

1 Introduction

In late 2024, the United States was in the midst of a presidential election when the decentralized prediction market, Polymarket [43], broke through mainstream news coverage [10,59]. Stories focused, in particular, on the fact that it offered odds more favourable to eventual winner Donald Trump than those reflected in conventional polls and forecasts. Polymarket's odds are not set by experts or pundits, instead it is a specific type of betting market where odds are extrapolated from the prices of trades made in an open market (or somewhat open, as Polymarket was banned at the time in many countries including the US).

As with traditional betting, whether online or through a bookie, prediction markets allow speculators to profit from correct forecasts [1,58]. However the structure of a prediction market is different than traditional betting. One key difference is that prediction markets ease the process of moving in and out of bets before the event resolves, encouraging traders to place bets if they think the odds are over- or under-stated, and withdrawing profits if the odds realign.

* A full version of this paper is available: https://arxiv.org/abs/2510.15612

It would be easy to think that Polymarket's design is the most obvious, straight-forward way to deploy a decentralized prediction market (DePM) on a blockchain. However the central thesis of this systemization of knowledge (SoK) paper is that Polymarket found success in bucking the trend. DePMs were first given a few paragraphs in the Ethereum vision paper [8], released in late 2013 for the blockchain that would be deployed in 2015. Then two 2014 papers presented flushed out systems: a whitepaper called Truthcoin [52] and an academic paper at WEIS 2014 [11] (informally known as the 'Princeton DePM' because of author affiliation). Developed independently,[3] the two papers' designs are vastly different, representing two different goalposts for how a DePM might look.

Early systems, like Augur [41] and Gnosis [23] closely resembled Truthcoin [52], while modern systems like Polymarket [43] either resemble the Princeton DePM [11] or use new solutions that resemble a hybrid of the two designs. Consider some examples:

1. From §3.3, in Truthcoin, the market creator is active in setting initial prices (*i.e.,* odds) for each option and risks its own money (bounded) [52]. In the Princeton system, the market creator is passive, not setting prices or risking any money [11]. Polymarket uses the latter [43].
2. From §3.5, in Truthcoin, outcome shares are created with predecessor to an automated market maker (AMM) [52]. In the Princeton DePM, outcome shares are traded with an orderbook [11]. Polymarket historically used an AMM and now uses an orderbook [43].
3. From §3.6, in Truthcoin, the blockchain decides event outcomes (*e.g.,* who won the election) through a reputation-based on-chain vote with slashing [52]. In the Princeton paper, they are resolved through trusted arbiters acting as oracles [11]. The Ethereum whitepaper suggests both [8]. In Polymarket, the third party oracle, UMA, operates under the hood through on-chain voting with slashing, but only when outcomes are disputed [43,56].

Research Questions. Noticing these points of differences inspired us to enumerate all of the design decisions involved in creating a DePM. What are the core modules that appear across DePM designs? How do DePMs differ in how they implement each module (creation, trading, resolution, settlement)? Which early design choices shaped later DePM architectures? Which design choices create recurring failure modes or operational problems? What trade-offs do DePMs make between decentralization, usability, and correctness of resolution? We also ask what open design gaps remain for building robust DePMs (these are marked as 'research gaps' in the paper).

Market Microstructure. In traditional financial markets, the mechanics of trading are called market microstructure (distinguished from the study of prices). Our SoK covers design options for different stages of DePMs but the most depth is on microstructure topics, including how shares are structured, traded, and settled.

[3] The Princeton paper describes Truthcoin as being released while the paper was under review [11], and the Truthcoin FAQ [51] mentions hearing about the Princeton paper but not having found the paper itself.

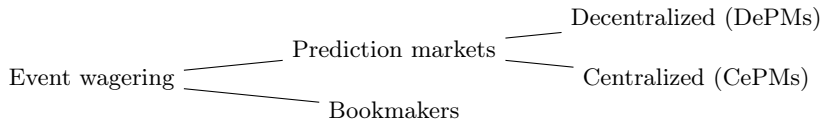
Methodology. We obtained a collection of academic works on decentralized prediction markets, as well as various intersecting topics including (centralized) prediction markets, oracles, DeFi, and AMMs. We used our knowledge of the field, Google Scholar (search and cited by features), and citations within papers. Our library is available, sorted by topic, on Zotero.[4] We also identified projects without white-papers or academic papers, reviewing the 97 current projects listed on a recently released online dashboard [49] and by studying 20+ historical systems that are no longer active. We also searched news sources for opinions and issues on leading decentralized prediction markets, such as Polymarket. We applied the following exclusion criteria to remove from consideration designs that are: fully permissioned CePMs, alternative wager structures beyond the first notable design we saw (*e.g.,* parimutuel betting [53,32,55], back/lay exchanges [12] and cash for difference contracts [16]), supporting/adjacent infrastructure, and insufficiently documented or scams. We reviewed over a hundred projects, distilled into the 35+ notable DePMs listed in the full paper.

A modular workflow (*cf.* [17]) is an abstraction that breaks competing designs into the same common set of steps and parts, enabling side-by-side comparison. We used multiple rounds of affinity diagramming to cluster DePM subcomponents from our corpus into eight stages: underlying infrastructure, market topic, share structure and pricing, trading, market resolution, settlement, and archiving. Relevant artifacts and data are available on GitHub.[5]

2 Preliminaries

2.1 Wagering Systems

Consider the following taxonomy of a few wagering systems:



Bookmakers versus prediction markets. A wager is a two-party contract with payouts based on the outcome of a future event. Consider Alice and Bob who wager on the same outcome of an event. With a fixed-odds bookmaker (or online betting), Alice's contract is different from Bob's contract in at least two regards: (i) it specifically names Alice as the counterparty and (ii) the payouts could be different if the odds changed between Alice's wager and Bob's. By contrast, in a prediction market contract (called a outcome share), Alice and Bob hold identical contracts: (i) all contracts are between the market operator and whoever redeems the contract, and (ii) the payout is exactly the same (typically $0 if

[4] Zotero: `https://www.zotero.org/groups/5750510/2024_prediction_markets/library`

[5] GitHub: `https://github.com/MadibaGroup/2024-PredictionMarkets`

incorrect and $1 if correct). Odds are reflected in the price paid for a prediction market contract (*i.e.,* variable cost and fixed payout), while a bookmaker contract has a fixed cost and variable payout. Thus the key distinction is that prediction market outcome shares are *fungible* and can be freely traded between participants, enabling a free market that communicates information to the public through outcome share prices, trading volume, market depth, and other financial market metrics. Operating a prediction market also differs from operating a bookmaker in a very important way:

Insight 1. Bookmakers quote odds without needing to hold sufficient funds in the event of a payout. It is merely a promise to pay. By contrast, a prediction market is structured with sufficient collateral for all outcomes at all times (including the very first trade! See §3.4). Every dollar paid to a winning trader is a dollar from either (a) a losing trader or (b) bounded within a special loss fund setup before the market opens.

CePM versus DePM. The term *decentralized prediction market* originates from the Ethereum whitepaper [8] and we abbreviate it DePM to match terms like DeFi (decentralized finance) [57] and DePIN (decentralized physical infrastructure networks) [34,37]. The term *decentralized* [40] in each of these is actually shorthand for both *decentralized* and *permissionless*, where permissionlessness is generally the more important way DePMs distinguish themselves from centralized prediction markets (CePMs). Permissionlessness could extend itself to the market topic, the trading of outcome shares, the closing of the market, and the withdrawing of rewards, but not all systems will open up each of these operations (as we will explain in §3). We say a system is DePM if at least one is permissionless.

2.2 Definitions

We start with an abstract definition of a DePM market, premised on outcome shares ($j \in J$) being represented as fungible token types with standard notions of accounts, transfers, minting (issuance), and burning (redemption). We do not try to capture CePMs designs that are margin-based (*e.g.,* Kalshi [29]) where traders hold only account-level positions, although such systems often satisfy analogous solvency and settlement conditions at the level of accounts rather than transferable tokens. Our definition generalizes on recent computer science-based definitions [5,19,48] which each presume specific sub-types of a prediction market (*e.g.,* [5,48] use *merging/splitting* while [19] presumes *automated bookmaking* from §3.5).

We distinguish an *event contract* from a *trading venue*. An event contract specifies the outcome space and the payoff rules for a fixed set of fungible token types. A trading venue (e.g., an order book or AMM) is an implementation choice that quotes prices and executes trades in some subset of those token types. Throughout, we use 'market' informally to refer to trading venues, and use 'event contract' for the event-level specification defined below.

Definition 1 (Event contract). *A (single) event contract is a tuple* $C = (\Omega, J, R, t_{\text{res}})$*, where* Ω *is a nonempty outcome space,* J *is a finite index set of fungible token types,* $R : \Omega \to \mathbb{R}^J_{\geq 0}$ *is a nonnegative payoff map, and* $t_{\text{res}} \in \mathbb{N}$ *is the (scheduled) resolution time. Intuitively,* C *is associated with an underlying uncertain event, whose possible resolutions are represented by* Ω*.*

Remark: WTA special-case. For an event contract $C = (\Omega, J, R, t_{\text{res}})$, suppose there exists a bijection $\iota : J \to \Omega$ and $R(\omega)_j \in \{0, 1\}$ with $\sum_{j \in J} R(\omega)_j = 1$ for all $\omega \in \Omega$. Then C is a winner-take-all (WTA) or Arrow–Debreu contract: a unit claim of token j pays 1 iff the realized outcome equals $\iota(j)$, and 0 otherwise.

Bundles and portfolios. A *bundle* is a specified collection of token types within an event contract (for example, a set of tokens that the system treats as a unit for minting, burning, or conversion). A *portfolio* is a vector $b = (b_j)_{j \in J} \in \mathbb{R}^J_{\geq 0}$ describing a trader's holdings, where b_j is the number of units of token type j. The payoff of a portfolio at outcome $\omega \in \Omega$ is $\langle b, R(\omega) \rangle = \sum_{j \in J} b_j \, R(\omega)_j$. Bundles are used to define mechanisms such as splitting/merging and other payoff-preserving conversions; portfolios are the positions held by traders.

Definition 2 (Prediction market system). *A prediction market system is a tuple* $\mathcal{S} = (\mathcal{C}, \mathcal{N}, \text{res})$*, where* $\mathcal{C}$ *is a countable set of event contracts,* $\mathcal{N}$ *is a numeraire (unit of accounting), and* res *is a resolution function over discrete time* $t \in \mathbb{N}$ *such that, for each* $C = (\Omega, J, R, t_{\text{res}}) \in \mathcal{C}$*, we have* $\text{res}(C, t) \in \{\perp\} \cup \Omega$ *for all* $t \in \mathbb{N}$*,* $\text{res}(C, t) = \perp$ *for* $t < t_{\text{res}}$*, and there exists* $\omega \in \Omega$ *such that* $\text{res}(C, t) = \omega$ *for all* $t \geq t_{\text{res}}$*.*

Definition 3 (System Axioms). *For every event contract* $C = (\Omega, J, R, t_{\text{res}}) \in \mathcal{C}$ *operating in system* $\mathcal{S} = (\mathcal{C}, \mathcal{N}, \text{res})$*, the following axioms hold. In what follows, all quantities are for this fixed event contract* C*, and we write* $\mathsf{Treas} := \mathsf{Treas}_C$ *and* $\mathrm{L} := \mathrm{L}_C$*.*

1. ***Issuance and Solvency.***
   Let $Q = (Q_j)_{j \in J} \in \mathbb{R}^J_{\geq 0}$ *be the vector of outstanding supplies, where* Q_j *is the total number of units of token* j *that have been issued and not yet burned, and let* $\mathsf{Treas} \geq 0$ *be the event contract's treasury (in the numeraire* $\mathcal{N}$*).*
   Define the worst-case liability

$$\mathrm{L}(Q) \;:=\; \sup_{\omega \in \Omega} \sum_{j \in J} Q_j \, R(\omega)_j. \tag{1}$$

   The system maintains, at all times,

$$\mathsf{Treas} \;\geq\; \mathrm{L}(Q). \tag{2}$$

   Any operation that changes supplies (including issuance) must preserve solvency. In particular, if issuance changes supplies from Q *to* Q'*, then the treasury is adjusted sufficiently:*

$$\Delta\mathsf{Treas} \;\geq\; \mathrm{L}(Q') - \mathrm{L}(Q). \tag{3}$$

Table 1. Classification of selected (for significance or novelty) DePMs using our modular workflow. Keywords used in the table are defined in referenced section, while ●, ◐, ○ respectively infer that the related property is fully, partially or not supported.

Year	System	Infrastucture (§3.1)	Permissionless Topic (§3.2)	Share Structure (§3.3)	Initial Issuance (§3.4)	Permissionless Trading (§3.5)	Resolution (§3.6)	Settlement (§3.7)	Arch. (§3.8)
2014	Truthcoin [52]	Sidechain	●	WTA	Bookmaking	●	Vote	Pull	●
2014	Princeton [11]	Sidechain	●	WTA	Splitting	●	Arbiter	Push	●
2015	Augur [41]	Smart Con.	●	WTA	Bookmaking	●	Vote	Pull	●
2019	Gnosis Sight [4]	Smart Con.	●	WTA	Splitting	●	Arbiter	Pull	◐
2019	Augur [42]	Smart Con.	●	WTA	Splitting	◐, ●	Vote	Pull	◐
2020	Polymarket [43]	Smart Con.	○	YNB(-NR)	Splitting	◐, ●	Opt., Vote	Gasless Pull	◐
2020	Omen [15]	Smart Con.	◐	YNB	Splitting	●	Opt., Vote	Push	◐
2021	Kalshi [30]	Centralized	○	YNB	Matching	○	Arbiter	Push	○
2022	Zeitgeist [60]	App-chain	◐	WTA	Bookmaking	○	Opt., Vote	Pull	◐

2. ***Transfer and Fungibility.*** *Holdings are transferable between accounts and transfers conserve per-token totals Q_j (the system-wide outstanding supply of token j).*
   Units of each token type are fungible: for all $j \in J$, any $\omega \in \Omega$, and any $a \geq 0$, a units of token j entitle the holder to $a\,R(\omega)_j$ units of $\mathcal{N}$ if the realized outcome is ω.
3. ***Settlement and Redemption.*** *If $t \geq t_{\text{res}}$ and $\text{res}(C, t) = \omega \in \Omega$, then any holder may redeem: for all $j \in J$ and $a \geq 0$, redeeming a units of token j pays $a\,R(\omega)_j$ units of $\mathcal{N}$. Redemption burns the redeemed tokens and debits the treasury by the corresponding payout, preserving the solvency invariant above.*

3 Modular Workflow

We now turn to the design landscape of DePMs and step through our modular workflow (*cf.* [17]) of eight stages: underlying infrastructure, market topic, share structure and pricing, trading, market resolution, settlement, and archiving. For each stage, we enumerate the possible designs and discuss competing trade-offs, summarized as the columns of Table 1.

Research Gap 1. Can accuracy metrics [22] be used to compare various CePM and DePM platforms? Are there meaningful correlations between accuracy and the market microstructure options in Table 1?

3.1 Underlying Infrastructure

Due to page restrictions, we relegate the designs for the underlying infrastructure to the full paper. It describes various blockchain options and notes that the

transparency of a blockchain can lead to copy-trading, posing a research gap related to that.

3.2 Market Topic

Due to page restrictions, we relegate the designs for the market topic to the full paper. It describes the differences between permissioned and permissionless systems for setting up what the market is about. Permissioned topic selection requires a central party, while permissionless designs can have problems with unethical markets (*e.g.,* assassination markets), spam, fragmentation, and poorly defined topics. The full paper also provides important real-world examples of ill-defined market topics that have proven to be problematic. It provides an insight and a research gap related to these points.

3.3 Share Structure and Pricing

The core requirement of a prediction market is that wagers are represented by fungible outcome shares. The structure of outcome shares typically falls into one of three categories and two variants (although more exotic structures are possible and explored in research).

WTA. The first structure we term *winner-take-all (WTA)* (*a.k.a. categorical*) and was popularized by Iowa Electronic Markets [27]. Consider a market with three possible outcomes: $\Omega = \{\mathsf{A}, \mathsf{B}, \mathsf{C}\}$. A WTA market issues an outcome share for each outcome $J = \{j_\mathsf{A}, j_\mathsf{B}, j_\mathsf{C}\}$. If there are only two outcomes, it is called a *binary market*. If the outcome ω is B, the share j_B pays \$1 (or one unit of numeraire $\mathcal{N}$) and the other shares pay \$0. For any $k \in \{\mathsf{A}, \mathsf{B}, \mathsf{C}\}$,

$$R(\omega)_{j_k} = \begin{cases} 1, & \text{if } \omega = k, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

For a WTA market to be well-functioning, two conditions must hold on outcome shares. (i) They should be *mutually exclusive* so no more than one share wins: $R(\omega)_{j_k} R(\omega)_{j_\ell} = 0 \quad \forall \omega \in \Omega, \forall k \neq \ell$; and (ii) they should be *complete* so at least one share wins: $\sum_{k \in \Omega} R(\omega)_{j_k} = 1 \quad \forall \omega \in \Omega$. If they are not mutually exclusive, the operator could be under-collateralized for making all payments (in violation of solvency in Definition 3). If they are incomplete, a deficient market might end with all participants receiving \$0. A consequence is that holding one share for each outcome is equivalent to holding \$1, a fact we will return to in §3.5.

In a WTA market, the price of a outcome share (*e.g.,* $p(j_\mathsf{A}) = \$0.54$) is a proxy for the probability that the outcome will occur (*e.g.,* $\Pr[\omega = \mathsf{A}] = 54\%$). A common adage is the prices of each share sum to \$1.00 ignoring fees and discounting (*e.g.,* $p(j_\mathsf{A}) = \$0.54$, $p(j_\mathsf{B}) = \$0.23$, $p(j_\mathsf{C}) = \$0.23$) but this is imprecise [11]. Outcome shares (like anything) have two prices: a bid price

(what a trader is willing to buy for) and an ask price (willing to sell for). If the sum of the bid prices exceeds \$1.00 or if the sum of ask prices are below \$1.00, arbitrageurs have an opportunity to secure risk-free profit through a trade that will erase the condition when fully extracted. This means the sum of bids and sum of asks should result in the bid-ask spread straddling \$1.00 but the amount of the spread could be arbitrarily large. So in user interfaces that display a single 'price' (*e.g.,* the last sale price or the midpoint between the best bid and the best ask), prices may indeed not sum to \$1.00—this is not a market failure, just a misunderstanding.

YNB. The second structure we term a *yes-no bundle (YNB).* YNB markets were popularized by InTrade [28]. A YNB market issues two token types for each outcome, a 'yes' and a 'no:' $J = \{j_{\mathsf{A_Y}}, j_{\mathsf{A_N}}, j_{\mathsf{B_Y}}, j_{\mathsf{B_N}}, j_{\mathsf{C_Y}}, j_{\mathsf{C_N}}\}$. For any $k \in \{\mathsf{A}, \mathsf{B}, \mathsf{C}\}$,

$$R(\omega)_{j_{\mathsf{k_Y}}} = \begin{cases} 1, & \text{if } \omega = k, \\ 0, & \text{otherwise.} \end{cases} \qquad R(\omega)_{j_{\mathsf{k_N}}} = \begin{cases} 1, & \text{if } \omega \neq k, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Each outcome-specific pair $\{j_{\mathsf{k_Y}}, j_{\mathsf{k_N}}\}$ constitutes a binary WTA market (k vs. not-k). A YNB market is the union of these pairs, so the WTA exclusivity and completeness properties hold per pair. However exclusivity and completeness do not necessarily hold across all bundles, allowing more flexible markets. For example, a market on what words Trump will say in a congressional address included Bitcoin (no), beautiful at least 10 times (yes), and Canada (yes).[6] Multiple words can resolve to yes (not exclusive) and it is possible he says none of the listed words (not complete). An established YNB market that is being actively traded, because it is not complete, can have new outcomes added to it fairly, whereas WTA markets must account for every possible outcome at the start of the market (perhaps utilizing an 'other' outcome).

YNB-NR. A variant of the YNB market is one where, even though it is not necessary, the yes share outcomes are in fact complete and exclusive. In other words, each yes/no bundle is a WTA market and the set of all yes shares is also a WTA market. We term this YNB variant as *negative risk* (YNB-NR), a term introduced by Polymarket [43]. Recall that in a WTA market, roughly speaking, the share prices sum to \$1 (modulo the fine print about bid/ask spreads above). For a YNB-NR market, the Yes shares sum to \$1, while the No shares will sum to $|\Omega| - 1$.

$$\sum_{k \in \Omega} R(\omega)_{j_{\mathsf{k_Y}}} = 1 \quad \text{and} \quad \sum_{k \in \Omega} R(\omega)_{j_{\mathsf{k_N}}} = |\Omega| - 1 \quad \text{for all } \omega \in \Omega. \tag{6}$$

Polymarket introduced a *negRisk* gadget that allows a trader holding a portfolio of No shares across A outcomes ($|A| > 0$) to extract cash by converting to the complimentary Yes shares:

[6] Polymarket: 'What will Trump say during address to Congress?'

$$\sum_{k \in A} R(\omega)_{j_{k_N}} = (|A| - 1) + \sum_{\ell \in \Omega \setminus A} R(\omega)_{j_{\ell_Y}} \quad \text{for all } \omega \in \Omega \text{ and any } A \subseteq \Omega. \quad (7)$$

Example. Alice owns 1000 No shares for Adam Back, Hal Finney, and Nick Szabo in a market about an HBO documentary set to identify Satoshi Nakamoto (see full paper). Since only one share type can win, Alice is guaranteed at least $2000 from her positions at resolution (and up to $3000 is she is right about all three). It is capital inefficient to keep her $2000 locked up in her portfolio of No shares. NegRisk allows her to converts her position into 1000 Yes shares for every other candidate and extract the $2000 in cash, maintaining the exact same exposure to the market ($1000 at risk).

Scalar. A third structure is provided in the full paper. It is not frequently used.

Research Gap 2. How does a *negRisk* gadget encourage price parity between equivalent positions? Can we quantify its improvements for price execution and trade execution costs? Does a reverse gadget, for converting a single yes share, offer similar benefits?

Research Gap 3. The academic literature has explored advanced market types, including combinatorial markets and establishing conditional probabilities from markets. Implementing and experimenting with these designs for DePMs is open. In sports-betting, chaining together multiple outcomes in an all-or-nothing bet (parlay) is popular but a treatment for DePMs is lacking.

3.4 Market Initialization

Options for trading outcome shares can be broken into two steps: (i) how does the first outcome share come into existence and how does the first trader trade, and (ii) how do traders trade once a market has been established? Probably the greatest evolution in DePMs, from Truthcoin to Polymarket, concerns how the first trade happens. There are three options: *automated bookmaking*, *splitting*, and *matching*.

Automated bookmaking was popularized through the academic work of Robin Hanson [25,26], researched for Intrade [36], and first suggested for DePMs by Truthcoin [52], which heavily influenced Augur v1 [41].

Definition 4 (Automated bookmaking). *Fix a market $M = (\Omega, J, R, t_{\text{res}})$. An* automated bookmaker *posts (i) a loss fund $\mathsf{Treas} \in \mathbb{R}_{\geq 0}$ in the numeraire $\mathcal{N}$ and (ii) a pricing rule (odds) for token types $j \in J$. It accepts deposits in $\mathcal{N}$ and issues shares, updating the supply vector S, and must always satisfy solvency (Definition 3).*

In this model, the operator sets initial prices for each outcome share (equivalent to setting market odds) and collateralizes enough payout money to cover

a worst-case loss in its treasury. If Alice is the first trader, she can immediately trade with the operator. The operator is autonomous and sets buy/sell prices algorithmically, originally using Hanson's logarithmic market scoring rule (LMSR) [26]. The key point is that the operator's fund is Alice's counterparty; if Alice wins, the fund loses, and vice-versa. The pros are instant liquidity for the first trader and the cons are the risk to the operator of losing money and the burden of needing to set initial odds (getting them wrong increases the chances it loses).

Acute readers might wonder the relationship between automated bookmaking and an automated market makers (*e.g.,* Uniswap), which are discussed in the next section. Roughly speaking, a WTA market run by automated bookmaking is termed a cost-function prediction market (CFPM) and a CFPM is equivalent in pricing (and trade costs) to an AMM (defined by a set of axioms) with the right invariant [19].

The second approach, *splitting*, was first suggested for DePMs by the Princeton DePM [11] based on the Iowa Electronic Market [27]. Augur switched from automated book making to splitting in v2 [42], Gnosis implemented splitting in CTF [13], and DePMs built on CTF, including Polymarket [43] and Omen [15], use it.

Definition 5 (Splitting and merging). *Fix a market* $M = (\Omega, J, R, t_{\text{res}})$ *in numeraire* $\mathcal{N}$*. A* splitting/merging *mechanism provides counterparty-free state transitions for a fixed* unit bundle B *of WTA token types such that*

$$\sum_{j \in B} R(\omega)_j = 1 \quad \textit{for all } \omega \in \Omega.$$

Splitting *maps* $1 \cdot \mathcal{N} \mapsto \sum_{j \in B} j$ *(minting one unit of each* $j \in B$*), and* merging *maps* $\sum_{j \in B} j \mapsto 1 \cdot \mathcal{N}$ *(burning them), typically before resolution. Repeating the operation yields multiple bundles.*

Recall that in a WTA market, exactly one share in a set of shares will payout \$1. This means that holding a complete portfolio of every share is equivalent (in payoff, ignoring fees and discounting) to holding \$1. A splitting gadget allows a trader to input \$1 and receive a portfolio contain 1 share of each outcome. Generally, a *merging* gadget is also available where a complete set can be redeemed for \$1 at any time before the market closes. Alice can obtain a set of shares and list asking prices (through a limit order book or by being the first liquidity provider in an AMM) for some or all of the shares, and if Bob is willing to buy a share from Alice, the first trade occurs. The pros to splitting is that the operator has zero exposure to the market, while the con is that Alice must wait for a second trader, Bob, before she can trade.

In YNB markets, each outcome's YES/NO pair is its own two-outcome WTA market. Splitting is per outcome: converting \$1 of collateral mints one $j_{\mathsf{k_Y}}$ and one $j_{\mathsf{k_N}}$ for the chosen k. Across outcomes, supplies are uncoupled—the total minted for the outcome A's bundle need not match that for outcomes B or C.

Insight 2. If the first trader makes one dollar, someone must lose a dollar. It either needs to be the operator or a second trader. This forces DePMs to choose between needing a loss fund or having no instant liquidity for the first trader.

The third approach, *matching*, was used by InTrade [28] and is used today by Kalshi [30]. We provide a description and research insight in the full paper.

3.5 Trading

Once outcome shares are in circulation, they can be traded any way fungible blockchain tokens can be traded. This includes *centralized exchanges* that custody the tokens and use central order book (CLOBs); *partially decentralized exchanges* where CLOB matching is done off-chain and settlement is done on-chain; or *fully on-chain exchanges*, of which automated market makers (AMMs) are the most common.

Of interest, AMMs were born out of Gnosis' research into automated bookmarking for prediction markets. They first developed alternatives to the LMSR rule, including the constant product rule [35]. In parallel, Bancor worked on exchanges where multiple traders could contribute tokens to a common liquidity pool. Uniswap v1 merged these two ideas to create the basic template of an AMM that is common today.

Circling back to automated bookmaking, splitting can be used in conjunction with AMMs to realize automated booking, as first used in Gnosis' closed beta Sight [4]. An operator begins with $2n$ units of the numeraire, *e.g.,* $2n$ USDC. It takes half of the money, n USDC, and uses the splitting gadget to obtain n shares in each outcome. For each outcome share, it estimates the outcome probability as p_i and starts an AMM with initial liquidity of n outcome shares and $p_i \cdot n$ USDC. It repeats for each outcome. Thus assuming the probabilities sum to 1, the operator has split its remaining n USDC across the AMMs. The operator can lose up to $2n$ USDC in divergence loss, assuming traders can split as well (and thus create the shares necessary to drain the AMM of its USDC), in addition to failing (through stale prices) to realize profits (*i.e.,* LVR [38]). The important point is that the operator is still solvent per Definition 3 as the $2n$ USDC is either locked in the splitting gadget or locked in the AMM.

Despite the direct lineage between prediction markets and AMMs, AMMs are problematic for prediction market trading. DePM outcome shares behave in specific ways that differ from typical crypto-assets and tokens. The price is strictly bounded between \$0 and \$1, the value of a share can jump to \$0 or \$1 near-instantly when an event outcome is finalized, and once finalized, the price is permanent. When real world events occur, AMMs can be drained faster than liquidity providers can withdraw liquidity.

Research Gap 4. Adapting AMMs to the unique constraints of predictions markets is at an early stage. Paradigm's pm-AMM tapers liquidity as a scheduled expiry approaches, which helps for events that crystallize over a fixed time horizon [39]; but many markets jump or resolve unexpectedly, falling outside of the model used for the results.

When trading on-chain, miners and other users can front-run transactions, an area of study called maximum extracted value (MEV) [14]. Although the term MEV did not exist at the time, the Princeton DePM describes the MEV problem extensively [11] and proposes a mitigation now called a frequent batch auction (FBA) (again, the term FBA was not popularized until later [7]).

Research Gap 5. Can we measurement MEV extraction for on-chain DePMs? What mechanisms for order-flow privacy apply to on-chain DePMs?

A final trading-relevant subject for prediction markets is arbitrage. Arbitrageurs ensure market prices are consistent, for example across all shares in a WTA market or between Yes/No bundles in a YNB market. A recent paper studies combinatorial arbitrage on Polymarket between markets with logically related predicates (*e.g.,* Republicans win the presidency; Trump wins the presidency) and measures roughly $40M USD of realized arbitrage profits over the measurement period [48].

3.6 Market Resolution

Market resolution approaches specify how a DePM determines and finalizes the winning outcome and triggers settlement. Designs vary in where the ground truth comes from (market prices, external sources, or participants), who can influence the decision, and what incentives/penalties secure it. In this section we review: oracle-less designs (on-chain truth, self-settling markets, auto-resolve rules), using a designated arbiter, using a network of reporters, and crowdsourcing a vote. The term *oracle* is commonly used for any of the latter three approaches.

Oracle-less. A market topic might reference *on-chain truth* (*e.g.,* total value locked in a DeFi service at the end of the year) in which case resolution can be made directly. A *self-settling market* [5] works with *splitting* and *merging* (see §3.5) of shares; it relies on participants with winning shares to purchase the losing shares (for close to $0) and redeem $1 by merging them. If losing shares do not trade near $0 or become illiquid, the broader market might accept winning shares as a substitute for dollars. If a market outcome is contentious with no recognized winner (*e.g.,* a poorly defined market in §3.2), the market will not settle. An *auto-resolve rule* could also be used such that outcomes are finalized when, say, their token trades above $0.99 for at least t time. Auto-resolve rules are subject to market manipulation (losing shares are wash traded near $1 long enough to finalize, in which case they are worth $1) or griefing attacks (yes shares are traded below $0.99 to prevent finalization). While an auto-resolve rule is too fragile to rely on solely, it could be used optimistically to provide an initial settlement; only disputed cases would be escalated to an oracle [2].

Designated arbiters. A CePM will decide events within the platform [29]. A DePM might outsource decisions to a trusted organization, such as the Associate Press (AP) which experimented with writing US election results into Ethereum.[7]

[7] Etherscan: AP U.S. Presidential Race Oracle

From a trust perspective, this is equivalent to using an AP CePM, but from a logistics perspective, it is simpler for AP to be an oracle than to run an entire CePM. Further, it allows *trust agility* where AP can be replaced by a competing source with minimal disruption [11]. A DePM might also used a designated party only as a last resort if other methods lead to a contentious result [47].

Recently interest is in evaluating arbitration through an AI system [9]. A 2025 Chainlink study poses 1,660 Polymarket questions to LLMs and reports 89.3% accuracy (sports 99.7%, politics 84.3%) with temporal-reasoning challenges [61]. AI can also be used in agent-mode, tasked with actively gathering evidence and reporting if resolution can be reached [9]. AI oracles are potentially vulnerable to adversarial manipulation and poisoning attacks [62], and may recall incorrect facts that are incorrectly stated in the training data. An over-reliance on AI systems in generating news, trading in markets based on news, and settling markets based on news could result in reflexivity and feedback loops.

Reporter networks. Some oracle systems focus on quickly reporting on updating values [6], such as an asset price or a sports score which could naturally settle some types of prediction markets, such as rapid markets from Augur Turbo [3] and Polymarket [44]. Generally a set (or network) of arbiters (or reporters) will report the value and an aggregation method (*e.g.,* median value or a mean value after outlier filtering for quantitative data, mode for qualitative) will update the value on-chain [17]. Updates are emitted when deviation or heartbeat thresholds are met, giving explicit control over latency/recency and bounding staleness under network stress. A DePM can read the aggregator at a fixed resolution time and settle without discretionary intervention [3,54].

Crowdsourced vote. A final mechanism is to determine off-chain facts by polling a set of willing participants [8,52,41,42,56,33]. This procedure is defined by four design choices:

1. *Electorate*: participants stake governance tokens to enrol and votes are either held with all participants, or with a randomly selected subset of all participants.
2. *Ballots:* may be visible to other participants or may be hidden until voting closes (typically with a two-round commit-reveal protocol but more sophisticated methods could be used [21]).
3. *Aggregation:* votes are weighed by the amount of staked tokens, categorical questions are decided by plurality (perhaps with minimum quorum) while quantities with a median or smoothed mean.
4. *Incentives:* participants who affirmed the final result in their individual votes receive rewards, while those who did not will at least forgo rewards, but might also have their participation tokens slashed.

The precise parameters of these systems vary significantly across many different designs, while deployed systems like UMA's data verification mechanism

(DMV) [56] and Kleros [33] have settled billions of dollars. Topics like fee mechanisms and proof of stake consensus protocols have received much academic attention, but incentive-compatible designs and formal proofs for oracles is largely under-explored. By contrast, a growing theoretical line proposes truth-elicitation schemes with formal guarantees [45,50]. However in practice, uptake is absent, in part because these designs are complex to run, sensitive to Sybil attacks, and can fail if voters rely on a common source of information.

Token votes for DePMs are hostile environments where an attacker's external payoff from a wrong resolution exceeds the system's economic security, which can lead to failure [18]. Rational voters are incentivized to agree with the majority, rather than necessarily reporting the ground truth (a situation colloquially known among economists as a Keynesian beauty contest). The system often works because the ground truth is an obvious point where participants will coordinate (known as a Schelling point), however voters might fear voting for truth if large token holders (whales) visibly stake on a wrong outcome early in the vote. An example[8] occurred on Polymarket [46], leading to concerns of token concentration within the UMA token. At the time of writing, the top five UMA wallets UMA control 45.6% of votes and the top 13 exceed 65%, enough to meet the passing threshold.[9] If successful attacks depress token prices, the attacks become cheaper, potentially creating a spiral (*cf.* [31]).

These concerns affirm design decisions like a secret ballot, using subsets of participants in a vote (potentially split by subject expertise), and redrawing larger panels on appeal with higher bonds/fees to raise capture costs only when disputes persist.

Research Gap 6. Deciding market outcomes is a uniquely DePM problem (since a CePM can arbitrate its own markets). Can we bring a theoretical lens to deployed oracle voting systems and understand what formal guarantees about incentives and safety can be made?

A DePM will probably use a *hybrid* or *chained approach* that does not rely on a single method. For example, an arbiter from an allowlist might be able to propose a market outcome while also allowing disputes from anyone. If an outcome is disputed, it is escalated to a crowdsourced vote. If the vote is considered defective, a further escalation could allow an Admin account to overrule the decision. The ultimate backstop is the law which would not stop a wrong outcome but could be used to remunerate parties damaged by it.

3.7 Settlement

Once the market is resolved, a DePM will enable each winning share to be converted into 1 unit of the numeraire (*e.g.,* 1 USD in a stablecoin) and transferred into the user's self-custody. While a DePM in theory could *push* payouts to users, it is common to wait for the user to initiate the redemption—a *pull*

[8] Polymarket: 'Ukraine agrees to Trump mineral deal before April?'
[9] Dune Analytics: $UMA Voting Participation

mechanism [13]. In this case, users pay the gas cost of the redemption which requires users to hold the native currency of the underlying blockchain. Polymarket offers gasless withdrawals (using OpenGSN relayers [24]), however users can bypass this at their choice. In a pull model, some users may not redeem their shares in a timely manner—a DePM may opt to sweep this surplus into its own capital or burn it, but DePMs generally hold it on-chain in perpetuity. As with any smart contract allowing withdraws based account balances, hardening against reentrancy attacks is critical.

The slogan 'sweat the game, not the payout [20]' is used to differentiate regulated sportsbook operators from 'neighbourhood bookies,' however even legitimate operators in the US have allegedly denied payouts using a legal loophole.[10] The advantage of a DePM in this context is two-fold: (i) payouts are fully (or largely) autonomous, not subject to human discretion, and (ii) the share structure ensures the operator has zero risk (or predetermined bounded risk, in the case of automated bookmaking) and is therefore financially indifferent to making any fair payout (see the solvency axiom in Definition 3).

Insight 3. A prediction market has no need to fight against paying out a bet the way bookmakers have in practice [20]. When operations are on-chain, solvency can be demonstrated directly from blockchain data.

3.8 Archiving

Due to page restrictions, we relegate the designs for archiving markets to the full paper. It describes how a market can be reconstructed after the fact, through a mix of on-chain data and off-chain data collected by the operator and possibly offered through APIs. It also presents a research gap related to this.

4 Conclusion

Researchers and builders have used a set of shifting designs, definitions, and vocabulary for DePMs. We aim to provide a modular framework that is useful for careful comparison between systems with different design choices, showcasing the full set of choices available, uncovering unsolved research problems, understanding the history of DePM ideas, and providing a learning resource for those wanting to catch up on DePMs. Our taxonomy does not identify a single best design but helps illustrate the trade-offs between them.

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[10] A law protects sports books from obvious errors, like a 'fat finger' when setting odds, but can be abused to claim long-shot bets were mistakes if they pay out [20].

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